

Gauranga Kumar Baishya

in LinkedIn — ✉ Email — ☎ Phone: +91 7086090441 — 🌐 GitHub — 🌐 Website —
🏠 Guwahati, Assam

EDUCATION

Grade 10 Shrimanta Shankar Academy	May 2016 CGPA: 10/10
High School, Grade 12 <i>Concentration: Mathematics, Physics, Chemistry, CS</i> Shrimanta Shankar Academy	Aug. 2017 – May 2019 CGPA: 8.76/10
B.Sc. Mathematics Tezpur University	Aug. 2019- Dec 2022 CGPA: 8.02/10
M.Sc. Mathematics (dropped out) Indian Institute of Technology Kharagpur	2022- 2023 CGPA: 8.75/10
M.Sc. Data Science Chennai Mathematical Institute	2023- present CGPA: 8.50/10

PUBLIC TALKS & EXPOSITORY RESEARCH PAPERS PRESENTED

Public Talk on Causal Inference An invited talk for the Google Development Group at IITM	Fall 2025 slides & more info
Public Talk on the Life and Work of John Von Neumann A talk for school kids on the occasion of National Math Day (22nd Dec)	Fall 2023 slides
Mathematics Olympiad Training Workshop/Camp A public talk arranged for state-level math olympiad winners	Autumn 2023 37th Annual Congress of AAM
Combinatorial proofs of two Euler type identities due to Andrews Cristina Ballantine, Richard Bielak For Pdf, click here	Spring 2021 For video, click here
The nil radical of a ring as an intersection of all the prime ideals of the ring Gauranga Kumar Baishya, Dr. Wade Bloomquist Poster & Presentation Slides	Spring 2021 For video, click here
The Gabriel's Horn Paradox The Presentation Nights, The Polymath Jr. REU For Pdf, click here	Spring 2021 For video, click here
End of program Presentation The Polymath Jr. REU	Spring 2021 For video, click here
Catalan Numbers A Public Talk	Spring 2021 For slides, click here
Generating Function for Bell Numbers A Public Talk	Spring 2021 For slides, click here

RESEARCH INTERNSHIP & PROJECTS

Summer Internship Harvard University, US (On Site) Harvard University, US	May 2024 - July 2024 Supervisor: Prof. Yu-Hua Tseng
• Deconvolution Analysis: Improved accuracy over an existing deconvolution tool, using a W-NNLS regression model to integrate single-cell RNA-seq data across multiple subjects, estimating cell type proportions in bulk RNA-seq while accounting for gene expression variability and cross-subject differences.	

Project | Normalized Cuts and Image Segmentation

Applied Linear Algebra project, Chennai Mathematical Institute (CMI)

August 2023 - November 2023

Supervisor: Dr. Priyavrat Deshpande

- Implemented graph-based image segmentation using normalized cuts (Shi and Malik), optimized via a generalized eigenvalue problem and spectral clustering. Efficiently segmented large-scale images by minimizing intra-group similarity and maximizing inter-group dissimilarity. Report

Project | Visualisation using R | Global Suicides (1985-2015) Dashboard

Institute of Mathematical Sciences, Chennai

August 2023 - November 2023

Shiny App link: Dashboard

- Developed a dashboard that provide some general insight on the delicate matter of suicides all over the world based on a dataset with data that range from 1985 to 2015 for every country accounting for sex, age and economic variables such as per capita GDP.

Project | Computational Genomics | Multimodal Single-Cell Integration

Institute of Mathematical Sciences, Chennai

April 2023 - October 2023

Supervisor: Dr. Sandeep Choubey

- The goal of the project is to predict how DNA, RNA, and protein measurements co-vary in single cells as bone marrow stem cells develop into more mature blood cells. I am trying to develop a model trained on a subset of a 300,000-cell time course dataset of CD34+ hematopoietic stem and progenitor cells (HSPC) from four human donors at five time points generated for this competition by Cellarity, a cell-centric drug creation company.

Reading internship | Theory Of Computation | CS

IIT Kharagpur

December 2022

Supervisor: Dr. Bodhayan Roy

- Studied Time and Space complexity, intractability and advanced topics which includes approximation algorithms, probabilistic algorithms, Alternation, Interactive proof systems, Parallel Computation and a bit of cryptography.

Reading Internship | Analytic & Algebraic Number TheoryFor certificate, click here

Summer 2021

Supervisor: Prof. Anupam Saikia, IIT Guwahati

- Did a rigorous reading and discussion of the book "A Classical Introduction to Modern Number Theory" by Ireland and Rosen.

Research in Computer Science | Combinatorics and CSPolymath Jr. REU

Summer 2021

Supervisor: Dr. Pat Devlin, Yale, US

- Did a research on the following problem: Suppose we pick positive integers n and k such that $n > k > 1$. The goal is to write down as many distinct permutations of Sym_n , the symmetric group of order n with the constraint that any two permutations that we pick must have a common subsequence of length at least k . What's the most number of permutations we could choose subject to this constraint?
- Did a reading of the book Erdős-Ko-Rado Theorems: Algebraic Approaches which describes the extremal combinatorics behind the research problem
- Presented a group talk titled "The Erdős Ko Rado group". Presentation link.

Use of Generating functions in Discrete Mathematics & CS | Combinatorics, CS

Cardiff University, Wales, UK

Spring 2021

Supervisor: Dr. Manjil Saikia

- Did a reading internship of the whole book, generatingfunctionology which studies some of the things like finding an exact formula for the members of a sequence, finding a recurrence formula, finding averages and other statistical properties of a sequence, finding an asymptotic formula for a sequence, proving unimodality, convexity etc and proving beautiful identities.

Study of Knot Theory and Abstract Algebra | Mathematics

Georgia Institute Of Technology, Atlanta, US

Fall 2020

Supervisor: Dr. Wade Bloomquist

- Did a reading internship Of Ring Theory; (Introduction to rings, Euclidean domains, Principal ideal domains, Unique factorisation domains and Polynomial Rings); Field Theory(Basic Theory of field extensions, Algebraic extensions, separable and inseparable extensions & cyclotomic polynomials and extensions.
- Did a reading internship of The Knot Book which describes the mathematical Theory of Knots.
- Prepared a small semester project titled "The nil radical of a ring as an intersection of all prime ideals."Poster & Presentation Slides.

Study of Representation Theory in the Symmetric Group | *Mathematics*

Cardiff University, Wales, UK

Fall 2020

Supervisor: Dr. Manjil Saikia

- Did a reading internship of the book, The Symmetric Book which studies group representations; matrix representations, G-modules and group algebra, complete reducibility and Maschke's theorem, G-homomorphisms and Schur's lemma, commutant and endomorphism algebras, group characters, inner products of characters, tensor products and restricted & induced representations.

Study of Partitions of numbers using Combinatorial techniques | *Combinatorics*

IIT-Delhi, India

Fall 2020

Supervisor: Dr. Biplab Basak

- Did a detailed study of the following: Let $a(n)$ be the number of partitions of n such that the set of even parts has exactly one element, $b(n)$ be the difference between the number of parts in all odd partitions of n and the number of parts in all distinct partitions of n , and $c(n)$ be the number of partitions of n in which exactly one part is repeated. Beck conjectured that $a(n) = b(n)$ and Andrews, using generating functions, proved that $a(n) = b(n) = c(n)$. A combinatorial proof of Andrew's result, it relies on bijections between a set and a multiset, where the partitions in the multiset are decorated with bit strings. Certificate.

MY PUBLISHED ARTICLES / WRITE-UPS1) **A very short anthology of some beautiful numbers, mathematical terms and expressions** [\[PDF\]](#)(It got published in the Gonitsora web magazine)2) **Catalan numbers and Dyck paths** [\[PDF\]](#)3) **Two advanced tests of convergence** [\[PDF\]](#)CONFERENCES, SEMINARS & WEBINARS ATTENDED**International Webinar titled "Galois groups of random integer polynomials"**

Prof. Manjul Bhargava, Princeton University

Spring 2021

International Webinar titled "Mathematics of the football"

Prof. Fernando R. Villegas, ICTP Trieste, Italy

Spring 2021

International Webinar (held via Zoom) titled "A brief history of Hecke operators"

Prof. Akshay Venkatesh, fields medalist, Institute for Advanced Study, Princeton

Spring 2021

International Webinar titled "The Riemann Hypothesis and why it is important"

Prof. Ken Ono, Thomas Jefferson Professor of Mathematics, University of Virginia

Spring 2021

International Webinar titled "Ramanujan Graphs and the Matrix Completion"

Prof. Mathukumalli Vidyasagar, FRS ; SERB National Science Chair and Professor, IIT Hyderabad

Spring 2021

International Webinar titled "The Geometry of a Robotic Arm in a Tunnel"

Dr. Cesar Ceballos, Technical University of Graz, Austria

Fall 2020

International Conference In Number Theory and Discrete Mathematics

American Mathematical Society

Fall 2020

International Conference In Cyber Security and Privacy

American Mathematical Society

Fall 2020

International Conference In Special Functions and Applications

BBD University, Lucknow, India

Fall 2020

International webinar in 1 & 2 dimensional holes in spaces

Prof. Matthias Kreck, University Of Bonn, Germany

Fall 2020

National Webinar on lengths of curves and Area of Figures

Prof. Rajeeva L. Karandikar, Ex director, Chennai Mathematical Institute (CMI), India

Spring 2020

National Webinar on The Congruent Number Problem & Elliptic Curves

Prof. Anupam Saikia, Indian Institute Of Technology (IIT), Guwahati, India

Spring 2020

International Webinar on Fermat's Last Theorem & a failure of Unique Factorization Dept. of Mathematics, IQAC Lumding College, India	Spring 2020
International Webinar on an Introduction to Algebraic Geometry Dr. Tarig Abdelgadir, Loughborough University, UK	Spring 2020

HONORS, FELLOWSHIPS AND AWARDS

Selected in the prestigious Khorana Scholars Program For link to the official site, click here.	March 2024
Secured an All Indian rank (AIR-89) in IIT-JAM in mathematics Scorecard	March 2022
Selected in Mathematics Training and Talent Search Programme (MTTS) Funded by NBHM, MTTS is the most popular national undergraduate/graduate training programme in Mathematics running in India. For link to the official site, click here.	June-July 2021
Madhava Mathematical Competition Special Recognition for securing Rank 1 in the entire North East Region, India.	Jan-2020
JNCASR Summer Research Fellowship A Highly Honoured Merit based grant for students pursuing education in STEM fields.	June 2020
An accomplishment in Quora My space in Quora earned 50k followers that discusses neuroscience & psychology	fall 2020-present
Best Academian Award A Trophy for excellence in academics during my high School education.	May 2017
Qualified NTSE (National Talent Search Examination) Scholarship. Merit based scholarship.	2017
Indian National Mathematics Olympiad (INMO) Among the top candidates in India, scoring appreciable marks in the Nationals.	2016
Regional Mathematics Olympiad (RMO) Qualified 5 times to the National Round.	2012-2017
Mathletics Competition Qualified 6 times to the State Merit List and secured State Rank 1 and 2 once.	2012-2016
State Math Olympiad Qualified it each time I appeared, into the State Merit List and secured State Rank 1 once.	2012-2016
State Chemistry Olympiad Qualified it 4 times, into the State Merit List (top 5), and secured State Rank 1 once.	2016
IASST National Science Day Quiz Won State rank 2 representing my school; prize distributed by Padma Shri awardee, Jadav Payeng.	2016
School Math Topper Award Secured School Rank 1 in International Mathematics Olympiad conducted by Science Olympiad Foundation.	2013
Intra school and Inter school Quizzes Participated and won many Co-Curricular Activity (CCA) Quiz competitions.	2012-2016

WORKSHOPS AND SUMMER SCHOOLS ATTENDED

AI and Machine Learning Summer School

Cambridge Centre for AI in medicine

September 2023

[exhibition](#)

- The CCAIM summer school in AI and ML in healthcare provided me with advanced insights into the application of sophisticated algorithms, enhancing my understanding of their intricate implementations and transformative impact in medical and healthcare research.

Applications of Machine Learning Techniques in Biology using Weka

Supervisor: Prof. M.M.Gromiha, IIT Madras

September 2022

[Certificate, click here](#)

- Learned bio-informatics, protein structure and function, protein interactions: computational techniques and handling computational biology lab. Used these knowledge to implement Machine Learning Techniques to understand protein structure and function, mutational analysis better.

Mathematics Training and Talent Search Programme (MTTS)-2021

[For link to the official site, click here](#)

12 June–4 July 2021

India

- Took advanced courses in Logic & Foundations of mathematics, Real Analysis, Linear Algebra, Geometry (curve tracing, sketching of surfaces, classification of quadric surfaces), Discrete Probability, Combinatorics and Elementary Number theory.

Lecture series on “Random Continued Fractions: A Markov Chain approach”

Prof. Alok Goswami, Indian Statistical Institute

April–May 2019

Kolkata, India

- Attended lectures on Markov Chains and processes and learned about Gauss Dynamical systems.
- Lectured on Advanced probability Theory and Graph Theory: Through Erdős Probabilistic solutions and random continued fractions.

Workshop on Mathematics Olympiad

Conducted by NBHM

2017

Assam, India

- Attended lecture series and problem solving sessions by the resource persons B.J.Venkatachala, Professor Of Mathematics, IISC Bangalore and national co-ordinator, Indian National Mathematics Olympiad (INMO), C.R. Pranesachar, professor HBCSE-TIFR, Prithwijit Dey (once deputy leader, Indian IMO team) and M.B.Rege, Regional Co-ordinator of Indian National Olympiads and Professor, dept of mathematics, NEHU.
- Attended various basic & advanced lectures on several topics of Number Theory, Geometry and Trigonometry, Combinatorics and Algebra.

Workshop on Physics and Astronomy, Biochemistry, Robotics and Electronics

Regional Science Centre

2015, 2016

Guwahati, Assam

- Attended lecture series and experimental sessions by Various professors from different universities on Robotics, Electronics, Physics and Astronomy and Biochemistry.

Workshop on Mathematics Olympiad Training conducted by NBHM

Gauhati University

2014

Guwahati, Assam

- Attended lecture series and problem-solving sessions by various professors from IITs & Guwahati University
- Attended various basic & advanced lectures on several topics of Number Theory, Geometry and Trigonometry, Combinatorics and Algebra.

INVOLVEMENT IN SOCIAL WORK

NSS

Tutored local underprivileged students in science and mathematics, a part of National Social Service(NSS).

Gonitsora

Presented papers and have been a constant in the weekly webinars of Gonitsora. Gonit Sora is a multi-lingual web magazine devoted to publishing well-written and original articles related to science and technology in general and mathematics in particular.

InnovateHer/Mathematical Enrichment Programme (Online Maths Training)

I established an initiative called “InnovateHer,” with a primary focus on promoting and popularizing mathematics education in the underserved areas of Assam and particularly among high school girls in the state through an another program named *Mathematical Enrichment Programme*, an initiative by AAM